

SANGUINE

DC2: Project Charter

Team 14A: Sanguine for Mayo Clinic - 13 January 2026

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Overview

Mayo Clinic's model of care emphasizes collaboration between multidisciplinary specialists, but external patients of the Mayo lab only get commoditized data. Mayo Clinic has an opportunity to scale its model of care to external patients and modernize test result delivery. Our system problem statement is **to** improve external patient care outcomes at scale **by** clearly, compassionately, and collaboratively augmenting the information provided with Mayo test results **using** modern technology, such as AI agents.

Strategic Business Context

Mayo has four strategic advantages and two strategic opportunities. Its advantages are its medical expertise in scientific accuracy, the breadth of tests available at its lab, its investments in data infrastructure, and its renowned brand. We have opportunities to improve the brand even further by modernizing its perception and to leapfrog competitors in terms of patient user experience.

Stakeholders

This table analyzes our stakeholders, categorizes them, and outlines their needs. This list is not exhaustive; we expect that the planning team will meet with these stakeholders to further detail their needs.

Category	Stakeholder	Needs
Primary beneficial	Patient	Fast and accurate test results, clear interpretation, understandable guidance, informed decision support, and competitive pricing.
Primary charitable	Primary Care Physician	Accurate and timely test results, searchable access through EMR, and in-depth clinical insights derived from test results.
Primary beneficial	Mayo Clinic Laboratories (MCL)	Visibility into clinical and patient context, operational efficiency, valid test samples, and appropriate data structures for test execution and result delivery.
Primary beneficial	Mayo Clinic Physicians	Access to patient history and clinical intent to support triage and interpretation before referral.
Primary beneficial	Mayo Clinic Leadership	Clear success metrics linked to brand and strategy, and a feasible, well-defined project plan.
Secondary problem	Mayo Clinic Legal	Documented proof of compliance and liability assessment.
Secondary beneficial	Software Developers (Mayo or External)	Clear software specifications and continuous feedback during development.
Secondary problem	Insurance providers	Price predictability, proof of diagnoses, and documentation of testing performed.
Secondary problem	Regulators (US and beyond)	Documented proof of regulatory compliance.
Secondary beneficial	Lab Technicians	Clear and unambiguous test execution instructions.
Secondary charitable	Researchers	Structured access to anonymized Mayo Clinic data.

PM Mandate

This charter serves as the mandate to the PM to evaluate the desirability and feasibility of augmenting existing technology to better serve external Mayo Clinic Laboratory patients.

- **Authorization:** The PM to utilize the allocated budget to plan the integration of technology to scale the Mayo Clinic Care Model for external patients. The PM is authorized to engage with stakeholders and coordinate with MITsdm Team 14A and Mayo Clinic's Dr. Chris Garcia, during the initial phase (January 2026 until May 2026).
- **Limitations:** The mandate is limited to feasibility planning. Execution of work breakdown structure and further resource allocation for implementation of this plan are outside the scope of this charter.
- **Deferred Investment:** Financial and operational commitment to implement this plan is deferred.

Opportunity

- **Strategic Context:** Leverage agentic AI to transform Mayo Clinic Laboratory results from commoditized data into a product that embodies the Mayo Clinic Model of Care (MCMC).
- **Direct Value:** Targeted application of specialty agentic AI can redefine the economic outlook of high quality post-analytical patient care. This enables the delivery of multi-disciplinary specialized expertise to external patients without the constraint of limited availability of Mayo Clinic physicians.
- **Indirect value:** Establish an enduring technology asset where physicians can seamlessly work alongside an agentic AI to scale external patient care. This platform enables improved post-analytical patient care outcomes while upholding compassion and collaboration, critical to the Mayo Clinic's global brand.
- **Positioning:** For Mayo Clinic Laboratory to maintain leadership of specialized “esoteric” testing by extending beyond a laboratory test provider into a dedicated partner in patient care. By investing in agentic AI capabilities that provide personalized, clinically sourced information, Mayo Clinic can leapfrog competitors who simply offer basic test results and non-specialized chatbots.

Targets and Tradeoffs

The planning team shall develop plans that meet the following target ranges:

1. Target Table

Dimension	Target (Desirable)	Acceptable Range	Notes
Scope / Performance	Feasibility assessment across 3 AI capability areas	All 3 required	EMR integration, analytics, communication
	Regulatory/compliance risk identification and mitigation	—	In the US, this means HIPAA, CLIA, and approval by FDA and CAP if necessary. Globally, others apply.
	Implementation roadmap with phasing options	Minimum 2 options	Build vs. buy analysis
	Applies to 5 different test types where it's needed most	4-6	“Metabolic panel” counts as one type, “blood cell count” is another, and so on.
Schedule	Planning complete by May 2026	Non-negotiable	MIT SDM program deadline
	Pilot program launched to trusted physicians only by September 2026	Anytime in September	To declare an initial launch in Q3.
	MVP launch: improving one test type for patients in one regulatory environment	January - February 2027	Launching anything in Q4 is hard, so we can wait until 2027.
	Global solution at scale	Q3 2027	To complete within fiscal year 2027
Cost	MIT SDM team effort ~200 hours	+20% flex	4 team members × ~50 hrs each
	Mayo staff time ~20 hours	+10 hours	Interviews + document review
	Capital expenditure \$0	\$0	Should leverage existing capital
	Software engineering cost (one-time)	\$10M - \$30M	Mayo would be happy with any budget within this wide range because cost is our lowest priority.
	Cloud expenses (monthly)	\$0.2M - \$0.5M	This is low because of Mayo's investments.

2. Priorities for Trade-off

Based on the strategic business context for this project, the following priorities and trade-offs are defined:

Priority: Scope → Schedule → Cost

PM to ensure comprehensive feasibility assessment takes precedence over timeline acceleration or effort minimization.

Tradeoffs:

Scope risk ↑ → simplify Could deliverables (competitive analysis, prototype concepts); preserve Must items (3 capability areas, compliance assessment, MCMC alignment)

Schedule risk ↑ → reduce Should items (early delivery, additional validation rounds); May 2026 deadline for planning completion remains non-negotiable

Cost risk ↑ → accept additional Mayo staff time if critical; team absorbs extra effort to protect scope completeness

Risks, Penalties, Constraints, and Bonuses

Type	Description
Risk	Any solution that gives interpretations or suggests actions to patients could risk malpractice liability.
Risk	A solution that's seen as low quality AI content could tarnish Mayo's brand of scientific excellence.
Risk	A solution that's seen as "replacing the physician", rather than empowering them, could make users turn against Mayo.
Risk	New regulation could obstruct any solution we build.
Penalty	A system that increases load on Mayo staff, either directly (via giving more tasks) or indirectly (by making patients ask more questions) would be penalized.
Penalty	Mayo has already made investments in AI, data centers, and technology. We should be able to leverage those; we would be penalized for requiring excessive investment.
Constraint	Any new insights would have to be delivered within the current EMR framework.
Constraint	Any implementation would have to stay within Mayo's technological strategy, like prioritizing GCP.
Constraint	We must comply with all laws and regulations, such as HIPAA.
Bonus	A system that can quantify its impact on external patient care outcomes would be a bonus. Normally, Mayo can't track what happens to an external patient after they get results, and this project doesn't have to change that, but it would help.
Bonus	It would be a bonus to be able to support many countries. To minimize initial scope, we will launch in only one country first.

Acknowledgement of AI Assistance

AI tools were used solely for editorial support, including improving clarity, conciseness, and organization of the written material. All analyses, assumptions, decisions, and conclusions are entirely the author's own.